

$^{36}\text{S}(\text{pol d}, ^3\text{He})$ 1985Kh04

$J^\pi=0^+$ for ^{36}S ground state.

1985Kh04: 52-MeV unpolarized and vector-polarized deuteron beams of 100 nA were produced from the Karlsruhe cyclotron. The target was a 1 mg/cm² ^{208}Pb sulfide with 81.1% enriched ^{36}S on ^{12}C backing. Reaction products were detected with two 300 μm and 1500 μm -thick $\Delta\text{E-E}$ surface-barrier detector telescopes (FWHM $\approx$ 90 keV). Measured angular distributions of differential cross sections $\sigma(\text{E}(^3\text{He}),\theta)$ and angular distributions of analyzing powers $i\text{T}_{11}(\text{E}(^3\text{He}),\theta)$. Deduced mass excess, levels, J , π , L-transfers, and spectroscopic factors from standard local, zero-range JULIE-DWBA analysis of $\sigma(\text{E}(^3\text{He}),\theta)$ and $i\text{T}_{11}(\text{E}(^3\text{He}),\theta)$.

 ^{35}P Levels

Spectroscopic factor $\text{C}^2\text{S}=\sigma(\theta)_{\text{exp}}/\sigma(\theta)_{\text{DWBA}}/\text{N}$, where $\text{N}=2.95$ is a normalization factor adopted by 1985Kh04 from 1974Ma34, originally from 1966Ba54.

$\text{E}(\text{level})^\dagger$	J^π	$\text{L}^\ddagger$	$\text{C}^2\text{S}^\ddagger$	Comments
0	$1/2^+$	0	1.63	
2386 6	$3/2^+^\#$	2	$0.31^\#$	
3857 2	$5/2^+@$	2	$2.91@$	
4474 21			<0.2	$1\text{d}_{5/2}$ proton transfer assumed in DWBA calculations, but 2016Mu03 proposed $1\text{f}_{7/2}$ ($\text{L}=3$) based on the measured parallel momentum distribution.
4665 3	$5/2^+@$	2	$1.06@$	
5189 13	$5/2^+@$	2	$1.38@$	
7520 30			<0.4	

† Deduced from ^3He spectra measured with unpolarized deuteron beam.

‡ From DWBA analysis of measured $\sigma(\theta)$.

$^\#$ L-1/2 transfer from analyzing power measurements; $1\text{d}_{3/2}$ proton transfer assumed in DWBA calculations.

$@$ L+1/2 transfer from analyzing power measurements; $1\text{d}_{5/2}$ proton transfer assumed in DWBA calculations.